

25) Find an equation involving g, h and K that makes this augmented matrix correspond to a consistent system:

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & K \end{bmatrix}$$

$$\text{Row 3} = \text{Row 3} + \text{Row 2}$$

$$\text{Row 3} = \text{Row 3} + 2 \times \text{Row 1}$$

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & 0 & 0 & h+K+2g \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & -3 & 5 & K+2g \end{bmatrix}$$

The corresponding equations are: $x_1 - 4x_2 + 7x_3 = g$
 $3x_2 - 5x_3 = h$
 $0 = h + K + 2g$

The augmented matrix correspond to a consistent system if and only if $(h + K + 2g = 0)$

26) Construct three different augmented matrices for linear systems whose solution set is $x_1 = -2, x_2 = 1, x_3 = 0$.

$$\text{Matrix 1: } \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Matrix 2: } \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Matrix 3: } \begin{bmatrix} 4 & 0 & 0 & -8 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The solution set $x_1 = -2, x_2 = 1$ and $x_3 = 0$ matches the solution sets of each of the matrices can be easily proved. $\square$

27) Suppose the system below is consistent for all possible values of f and g . What can you say about the coefficients c and d ? Justify your answer.

$$\begin{cases} 4x_1 + 3x_2 = f \\ cx_1 + dx_2 = g \end{cases}$$

Soln: Augmented matrix: $\begin{bmatrix} 1 & 3 & f \\ c & d & g \end{bmatrix}$

$$\text{Row 2} = \text{Row 2} - c \times \text{Row 1}$$